

CO 250 Spring 2026: Assignment 5 solutions

- A5-1. (a) i. Let e_i be the elementary vector of all 0's except 1 at entry i . Suppose the columns of $A_{B'}$ is the set $C_{B'} = \{e_1, \dots, e_n, A_k\} \setminus \{e_\ell\}$. The column $A_k = (A_{1k}, \dots, A_{nk})^T$.
 $(\Rightarrow)$ Suppose that the ℓ -entry of A_k is 0, i.e. $A_{\ell k} = 0$. Then $A_k = A_{1k}e_1 + A_{2k}e_2 + \dots + A_{nk}e_n$ (note that this combination does not include e_ℓ since $A_{\ell k} = 0$). Therefore, $C_{B'}$ is not linearly independent.
 $(\Leftarrow)$ Suppose that the ℓ -entry of A_k is not 0. Consider any linear combination of $C_{B'}$ that equals 0. Since A_k is the only vector in $C_{B'}$ where the ℓ 's entry is not zero, the coefficient of A_k in the linear combination is 0. Disregarding A_k , we see that each e_i is the only remaining vector in $C_{B'}$ where the i -th entry is not zero, so all of the coefficients must be 0. Hence $C_{B'}$ is linearly independent.
- ii. In simplex, once we have selected entering variable k , then we need to check t by looking at $x_B = b - tA_k$. We choose $t = \min\{\frac{b_i}{A_{ik}} : A_{ik} > 0\}$. Since we have successfully chosen a leaving variable ℓ , this means that the prerequisite is $A_{\ell k} > 0$. Hence the ℓ -th entry of A_k is not 0.
- (b) The optimal solution for our basis is $(0, 0, 0, 0, 2, 5, 0)^T$ with optimal value 15. If there is another optimal solution x , then its objective value is also 15, so $(-3, 0, -1, -4, 0, 0, 0)x = 0$. But $x \geq 0$, so we must have $x_1 = x_3 = x_4 = 0$. This leaves us with 4 variables that satisfy the following constraints:

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 2 & 0 & 1 & 0 \\ -3 & 0 & 0 & 1 \end{pmatrix} x = \begin{pmatrix} 2 \\ 5 \\ 0 \end{pmatrix}$$

$$x \geq 0$$

Solving this using linear algebra, we use a parameter t to represent x_2 , which gives us

$$x_5 = 2 - t, \quad x_6 = 5 - 2t, \quad x_7 = 3t.$$

We need non-negativity, so $0 \leq t \leq 2$. We now define

$$x(t) = \begin{pmatrix} 0 \\ t \\ 0 \\ 0 \\ 2 - t \\ 5 - 2t \\ 3t \end{pmatrix}.$$

For any $0 \leq t \leq 2$, $x(t)$ is feasible. Also, $(-3, 0, -1, -4, 0, 0, 0)x(t) = 0$, so $x(t)$ has objective value 15. Therefore, $x(t)$ is optimal for all $0 \leq t \leq 2$, and these are all the optimal solutions.

A5-2. (a) We first add one auxiliary for each constraint to get

$$\begin{pmatrix} -1 & -2 & 1 & 1 & -2 & 1 & 0 \\ 2 & 4 & -1 & -4 & 3 & 0 & 1 \end{pmatrix} x = \begin{pmatrix} 5 \\ 5 \end{pmatrix}.$$

The objective function is the sum of the negatives of the auxiliary variables.

$$\max \quad (0 \ 0 \ 0 \ 0 \ 0 \ -1 \ -1).$$

In canonical form, the auxiliary LP we use to start phase 1 is

$$\begin{aligned} \max \quad & (1 \ 2 \ 0 \ -3 \ 1 \ 0 \ 0) - 10 \\ \text{s.t.} \quad & \begin{pmatrix} -1 & -2 & 1 & 1 & -2 & 1 & 0 \\ 2 & 4 & -1 & -4 & 3 & 0 & 1 \end{pmatrix} x = \begin{pmatrix} 5 \\ 5 \end{pmatrix} \\ & x \geq 0 \end{aligned}$$

Solving this using Simplex, we get the following optimal canonical form.

$$\begin{aligned} \max \quad & (0 \ 0 \ 0 \ 0 \ 0 \ -1 \ -1) \\ \text{s.t.} \quad & \begin{pmatrix} 1 & 2 & 0 & -3 & 1 & 1 & 1 \\ 0 & 0 & 1 & -2 & -1 & 2 & 1 \end{pmatrix} x = \begin{pmatrix} 10 \\ 15 \end{pmatrix} \\ & x \geq 0 \end{aligned}$$

Since the objective value is 0, we have found a feasible solution to the original LP $(10, 0, 15, 0)^T$ with basis $\{1, 3\}$. We can find the canonical form for this basis from the original LP to start phase 2.

$$\begin{aligned} \max \quad & (0 \ -1 \ 0 \ 4 \ 3) + 75 \\ \text{s.t.} \quad & \begin{pmatrix} 1 & 2 & 0 & -3 & 1 \\ 0 & 0 & 1 & -2 & -1 \end{pmatrix} x = \begin{pmatrix} 10 \\ 15 \end{pmatrix} \\ & x \geq 0 \end{aligned}$$

We solve this using Simplex. But we notice that the coefficient of x_4 in the objective is positive, while the column of x_4 consists of only negative numbers. This implies that the LP is unbounded. We set $x_4 = t$, and keep the other non-basic variables $x_2 = x_5 = 0$. We get $x_1 = 10 + 3t$ and $x_3 = 15 + 2t$. So a series of feasible solutions whose objective values approach infinity is $(10 + 3t, 0, 15 + 2t, t, 0)^T$ for $t \geq 0$. A certificate of unboundedness is then $\bar{x} = (10, 0, 15, 0, 0)^T$ and $d = (3, 0, 2, 1, 0)^T$.

(b) We first multiply the second row by -1 and add one auxiliary variable for each constraint to get

$$\begin{pmatrix} 2 & 1 & 4 & -2 & 1 & 1 & 0 \\ 1 & -1 & 3 & -3 & 0 & 0 & 1 \end{pmatrix} x = \begin{pmatrix} 3 \\ 3 \end{pmatrix}$$

The objective function is the sum of the negatives of the auxiliary variables.

$$\max \quad (0 \ 0 \ 0 \ 0 \ 0 \ -1 \ -1).$$

In canonical form, the auxiliary LP we use to start phase 1 is

$$\begin{aligned} \max \quad & (3 \ 0 \ 7 \ -5 \ 1 \ 0 \ 0) - 6 \\ \text{s.t.} \quad & \begin{pmatrix} 2 & 1 & 4 & -2 & 1 & 1 & 0 \\ 1 & -1 & 3 & -3 & 0 & 0 & 1 \end{pmatrix} x = \begin{pmatrix} 3 \\ 3 \end{pmatrix} \\ & x \geq 0 \end{aligned}$$

Solving this using Simplex, we get the following optimal canonical form.

$$\begin{aligned} \max \quad & (-1/2 \ -7/4 \ 0 \ -3/2 \ -3/4 \ -7/4 \ 0) - 3/4 \\ \text{s.t.} \quad & \begin{pmatrix} 1/2 & 1/4 & 1 & -1/2 & 1/4 & 1/4 & 0 \\ -1/2 & -7/4 & 0 & -3/2 & -3/4 & -3/4 & 1 \end{pmatrix} x = \begin{pmatrix} 3/4 \\ 3/4 \end{pmatrix} \\ & x \geq 0 \end{aligned}$$

We see that the optimal value is negative, meaning it is impossible to have a feasible solution where all auxiliary variables are 0's. Therefore, the original LP is infeasible.

The final basis is $B = \{3, 7\}$. From class, we calculate the certificate of infeasibility by

$$y = A_B^{-T} c_B = \begin{pmatrix} 4 & 0 \\ 3 & 1 \end{pmatrix}^{-T} \begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 3/4 \\ -1 \end{pmatrix}.$$

But when we obtained the auxiliary LP, we multiplied the second row by -1 . So we need to reverse this by multiplying the second entry of y by -1 as well. So a certificate of infeasibility is $y = (3/4, 1)^T$.

We can verify that this is the correct certificate.

$$\begin{aligned} y^T A &= (3/4 \ 1) \begin{pmatrix} 2 & 1 & 4 & -2 & 1 \\ -1 & 1 & -3 & 3 & 0 \end{pmatrix} = (1/2 \ 7/4 \ 0 \ 3/2 \ 3/4) \geq 0^T \\ y^T b &= (3/4 \ 1) \begin{pmatrix} 3 \\ -3 \end{pmatrix} = -3/4 < 0 \end{aligned}$$

So our certificate is correct.

- A5-3. (a) The objective value of $\begin{pmatrix} \bar{x} \\ \bar{x}' \end{pmatrix}$ is $-\mathbb{1}^T \bar{x}'$. This is 0 by assumption. But $\bar{x}' \geq 0$, so $-\mathbb{1}^T \bar{x}' \leq 0$. Therefore, it must be the case that $\bar{x}' = 0$.

Substitute into the constraints of (AuxP), we get $A\bar{x} + I\bar{x}' = b$, which implies that $A\bar{x} = b$. Together with $\bar{x} \geq 0$, we see that $\bar{x}$ is a feasible solution to (P).

- (b) We claim that the objective function of (AuxPOpt) is $-\mathbb{1}^T x'$.

Note that we can rewrite the objective function of (AuxP) as $(0 \ -\mathbb{1})(x \ x')^T$, and the equality constraints as $(A \ I)(x \ x')^T = b$. The formula for the objective function of the canonical form with respect to basis B is then

$$[(0 \ -\mathbb{1}) - y^T(A \ I)](x \ x')^T + b^T y$$

where

$$y = A_B^{-T} c_B.$$

Since $B \subseteq \{1, \dots, n\}$, $c_B = 0$. So $y = 0$. Hence the objective function of (AuxPOpt) is

$$(0 \quad -\mathbb{1})(x \quad x')^T = -\mathbb{1}x'.$$

- (c) Since B is a basis for (AuxP), there are m elements in B . But B' is smaller than B , so it has fewer than m elements. Hence B' cannot be a basis for (P).
- (d) We see that B' corresponds to a subset of columns of A that do not span $\mathbb{R}^m$. Using a result from linear algebra about extending a basis, we can add additional columns of A to form a basis $\bar{B}$ that span $\mathbb{R}^m$. So then $\bar{B}$ is a basis that contains all elements of B' . From part (a), we know that $\bar{x}$ is feasible for (P). Also, since $\bar{x}$ is a BFS for the basis B , only entries of B' could be non-zero. Since $\bar{B}$ contains all of B' , all entries not in $\bar{B}$ are 0's. Hence it is a BFS for $\bar{B}$.